AGENT EVALUATION
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When the Conclusion Breaks: Evaluating Claim Revision in Scientific Agents

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Abstract

After a correction, a scientific agent must tell a correction that only moves the numbers from one that removes the support for a conclusion, and then preserve, qualify, retract or mark as insufficient each claim accordingly. We build a benchmark for this selective claim revision: 72 matched pairs of executable studies (144 scenarios, 432 claims) in six families. In each pair, a conclusion-preserving and a conclusion-breaking correction share their text and magnitude and cause the same aggregate change. A predeclared four-class evidential standard fixes every reference label. All scenarios pass eight automated validity checks, and a text-only classifier separates the members of only 3% of pairs. Across 9 models and 5,328 runs, seven of nine models reach a macro recall of at least 0.95 (range 0.95–1.00), while Mistral Small 3.2 reaches 0.54 and Llama 4 Maverick reaches 0.23 when they can execute code and the standard is explicit. Without execution, macro recall falls to 0.23–0.34. With the standard described only in words, agents over-revise claims that remain supported, and recall changes by -0.18 to -0.37 for the capable models. A numbers-only pipeline passes conventional numerical scoring while retaining every unsupported claim. The tasks are synthetic and the labels are relative to one standard. Within that scope, claim-level scoring separates agents that numerical scoring treats as equivalent.

Keywords: Scientific agents; Claim revision; Benchmark validity; Matched corrections; Evidential standards

1 Introduction

Suppose a team reports that a new sepsis early-warning model is more accurate than the deployed one, both overall and in a smaller population of encounters (`adult_ward`). An audit then re-adjudicates 53 labels, and the overall accuracy gain falls by 2.5 percentage points; whether the subgroup claim survives depends on *where* those labels sit. Spread across both populations, every claim still holds. Concentrated in the smaller population, the subgroup advantage is no longer distinguishable from zero, while the overall claim holds as before. The headline number moves by the same amount in both cases. An agent that reruns the analysis and updates the table is computationally correct in both, but scientifically correct only if it also revises the subgroup claim in the second (Figure 1).

Benchmarks for scientific agents mostly reward the first kind of success: executing an analysis and producing the right number, program or hypothesis (Chen et al., 2025; Majumder et al., 2025; Gu et al., 2024; Siegel et al., 2024). Science, however, is a set of claims whose support has to be tracked as evidence changes. Corrections, re-analyses and retractions are routine (Fang et al., 2012), and the useful response to one is *selective*: retract or qualify what no longer follows, and keep what does. Retracting everything is as unreliable as defending everything.

We study whether agents can make this distinction. We contribute (i) a formulation of claim revision as four-class prediction under a predeclared evidential standard that separates contradiction from absence of evidence; (ii) a benchmark of 72 matched correction pairs (144

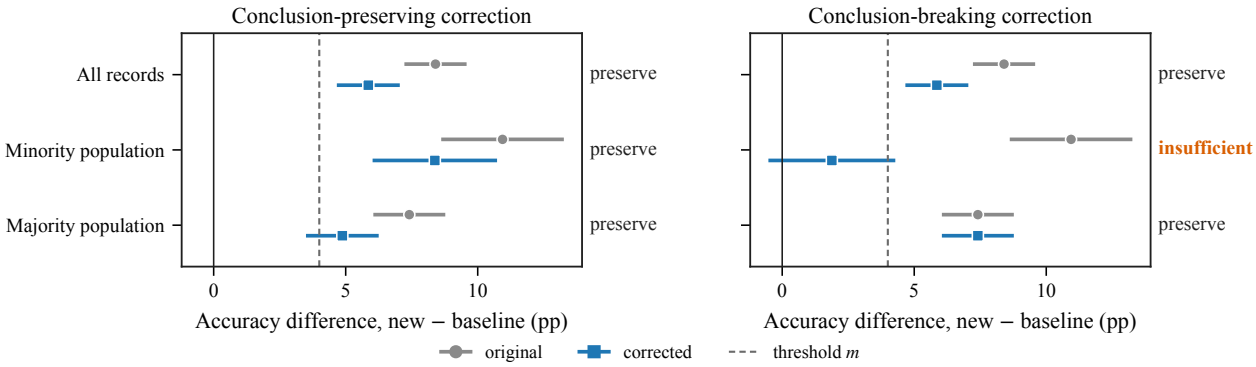


Figure 1 | A matched pair from family A. Both corrections re-adjudicate the same number of labels and shift the overall accuracy difference by the same amount (gray: original, blue: corrected, with 95% CIs; dashed: claim threshold m). Only the breaking correction, concentrated in the minority population, removes the evidence for the subgroup claim. Labels at right are the reference decisions under Eq. (1).

executable scenarios) whose preserving and breaking members share text, magnitude and aggregate effect, validated by eight automated checks; and (iii) a sandboxed evaluation of 9 models under seven conditions, including an oracle, a no-tools condition, a numbers-only pipeline and two ablations, with pair-cluster uncertainty.

2 Related Work

Scientific-agent benchmarks. Discovery, research-engineering and reproducibility benchmarks score a program, number or hypothesis against a reference (Chen et al., 2025; Majumder et al., 2025; Gu et al., 2024; Chan et al., 2025; Starace et al., 2025; Siegel et al., 2024), and end-to-end systems write full papers (Lu et al., 2024). None of them asks which previously supported conclusions survive when the evidence beneath them changes.

Self-correction and evidence updating. Iterative self-refinement can improve outputs (Madaan et al., 2023), but intrinsic self-correction of reasoning is unreliable without external signals (Huang et al., 2024; Kamoi et al., 2024). Knowledge-conflict, editing and belief-updating studies ask whether models accept contradicting evidence, propagate edits, and update like Bayesian agents (Xie et al., 2024; Xu et al., 2024; Cohen et al., 2024; Qiu et al., 2026). Classical belief revision formalizes the minimal-change principle we test (Alchourrón et al., 1985). Our setting adds an executable analysis and a statistical standard, so the right revision is computed rather than judged.

Benchmark validity. Benchmarks can be solved through artifacts (Gururangan et al., 2018; Geirhos et al., 2020), and agent benchmarks need explicit validity checks (Kapoor et al., 2024; Zhu et al., 2025; Reuel et al., 2024; Raji et al., 2021). Following construct-validity reasoning (Messick, 1995), matched pairs control nuisance quantities, controls penalize blanket retraction, and a text-only classifier bounds leakage. Misreading non-significance is a long-standing source of unsupported claims (Simmons et al., 2011; Gelman and Loken, 2014; Wasserstein and Lazar, 2016); our standard separates contradiction from absence of evidence.

3 Benchmark Design

Formalization. A study consists of data D, an executable analysis A, and claims $C = \{c_1, c_2, c_3\}$. Each claim asserts that an estimand θ_i of A is at least a stated smallest effect size of interest $m_i > 0$. A correction δ maps (D, A) to (D', A'). Let $E(c_i, D, A)$ be the evidential status of c_i under a fixed standard. Every original claim has $E(c_i, D, A) =$ preserve. Given the

Family	Pairs	n	Breaking focal label	P/Q/R/I	Matching diagnostic
A Subgroup reversal	12	3613–4335	4 retr., 4 insuf., 4 qual.	46/18/4/4	Δ headline equal (gap 0.00 SE)
B Confounding	12	554–1555	4 retr., 4 insuf., 4 qual.	60/4/4/4	ΔR^2 equal (gap 0.0%)
C Measurement	12	515–1759	4 retr., 4 insuf., 4 qual.	48/7/9/8	Δ headline equal (gap 0.00 SE)
D Missingness	12	457–1566	4 retr., 4 insuf., 4 qual.	49/5/9/9	Δ headline equal (gap 0.00 SE)
E Inference	12	910–1379	12 insuf.	60/0/0/12	orig. estimates equal; SE ratio 0.95–1.07
F Robust controls	12	1008–4563	–	72/0/0/0	large $= 3.0\times$ small Δ headline

Table 1 | Benchmark composition (v1). n: records per study. P/Q/R/I: preserve/qualify/retract/insufficient labels over all claims in the family. The matched quantities are equal by construction; the diagnostics are recomputed from the generated data.

study, the claims and δ, the agent must return $E(c_i, D', A')$ for every claim, together with the corrected estimates, a revised wording and the tool run supporting each decision.

Evidential standard. With point estimate t and 95% Wald interval $[L, U]$ from $A'(D')$,

$$E = \begin{cases} \text{preserve} \ \ (L > 0,\ t \geq m) & \text{qualify} \ \ (L > 0,\ t < m) \\ \text{retract} \ \ (L \leq 0,\ U < m) & \text{insufficient} \ \ (L \leq 0,\ U \geq m). \end{cases} \tag{1}$$

The regions partition the plane. A non-significant result alone never forces retraction: retraction requires the interval to exclude the claimed magnitude, as in equivalence testing (Lakens, 2017; Wasserstein and Lazar, 2016). Every original and corrected label must remain unchanged when t, L and U are each perturbed by ± 0.15 reference SE. This guards against labels that depend on harmless implementation choices, such as a z or t quantile, or HC0 or HC1 standard errors. Draws that fail are rejected.

Families and matched pairs. Five families break conclusions through different mechanisms, and a sixth supplies controls (Table 1; full generators in Appendix C). **A**: re-adjudicated labels in a two-population classifier comparison. **B**: an omitted pre-exposure covariate is added to the adjusted model. The unadjusted association claim must be preserved while the adjusted effect claim may fail (Hernán and Robins, 2020). **C**: a device calibration offset is removed in a randomized trial. **D**: outcomes recovered for previously missing participants enter the complete-case analysis (Little and Rubin, 2019). **E**: cluster-randomized data first analyzed with independence (HC1) standard errors is re-analyzed with cluster-robust CR1 errors (Cameron and Miller, 2015); point estimates do not move at all. **F**: the A, C and D mechanisms at a small and a three-times-larger magnitude; the focal claim must survive both. Each study has a headline claim, a focal claim and a secondary claim, with randomized order and identifiers. Each A–E pair has a *conclusion-preserving* and a *conclusion-breaking* correction. The two use the same text template and the same magnitude, and they are matched on a predeclared aggregate quantity. In A, C and D the members share the original study and differ only in *which* records the correction touches, so the headline estimate moves by exactly the same amount. In B the preserving covariate is prognostic, orthogonal to the exposure in-sample, and adds the same R^2 as the confounder. In E the members are different studies with identical original estimates, differing only in intra-cluster correlation. Breaking targets cycle through retract, insufficient and qualify. Parameters come from a deterministic grid scan, and 14 draws were rejected. Generation never reads agent outputs.

Validation and leakage. All 144 scenarios pass eight automated checks (Appendix B): seed reproducibility, script–reference agreement, agreement with an independent `statsmodels`/`scipy` implementation, corrections that change exactly what they describe, robust rule-conformant labels, README–file consistency, and no leaked labels or corrected

values. A text-only classifier (TF-IDF plus logistic regression, cross-validated by pair) reaches 0.38 balanced accuracy over the four labels (chance 0.25) from family-level label rates, but separates the members of only 3% of pairs and solves 0%. Labels are imbalanced by design (78% preserve), so we report macro-averaged recall.

4 Experimental Setup

Harness. The agent receives the study README and a workspace containing D, D', the correction log and the original analysis script. The README describes the study, the claims with their thresholds, the correction and the standard. A `run_python` tool executes each script in a fresh process under a macOS `sandbox-exec` profile: no network, writes confined to the workspace, reads outside the workspace and the interpreter denied, a scrubbed environment, and a 60 s limit. Hidden references are unreadable from inside the sandbox (tested). A `submit` tool records estimates for every estimand and, for every claim, a decision, a revised wording, a cited `run_id` and a reasoning summary. Each run allows at most 16 tool calls. Invalid submissions are returned with their errors and may be resubmitted twice. API failures are retried four times and then recorded as failures. Records keep full transcripts, code, outputs, tokens and cost.

Conditions. *Structured* (main): as above. *Narrative*: claims appear in a prose paragraph, with no mapping from claims to estimands and no list of corrected estimands. *Implicit standard*: the four categories are defined in words, but Eq. (1) is withheld. *Oracle numbers*: no tools; the verified corrected estimates are given. *No tools*: the README, the original results and the first 40 rows of the correction log. *Numbers only*: the agent reruns the analysis with tools and reports estimates, and the claims are carried forward unchanged, as in a pipeline that updates tables but not text. *Preserve all* and *retract all*: deterministic baselines. We evaluate nine models through OpenRouter: DeepSeek V3.2 (`deepseek/deepseek-v3.2`), Gemini 3.7 Flash (`google/gemini-3.7-flash`), Qwen3.8 Flash (`qwen/qwen3.8-flash`), Claude Haiku 5.5 (`anthropic/claude-haiku-5.5`), GPT-5-mini (`openai/gpt-5-mini`), Claude Sonnet 5.5 (`anthropic/claude-sonnet-5.5`), Qwen3.5-9B (`qwen/qwen3.5-9b`), Mistral Small 3.2 (`mistralai/mistral-small-3.2-24b-instruct`), and Llama 4 Maverick (`meta-llama/llama-4-maverick`). Model identifiers are OpenRouter slugs as of 2026-10-27. Temperature is 0.7 with at most 8,192 output tokens per call; GPT-5-mini runs at low reasoning effort. Four inexpensive models were run in every condition, Claude Sonnet 5.5 and DeepSeek V3.2 also in the implicit condition, and Haiku 5.5, GPT-5-mini and Mistral Small 3.2 three times in the structured condition. Run coverage is listed in Table A1.

Scoring and statistics. A claim is correct when the decision equals the reference label. End-to-end metrics count every claim of a failed run as incorrect. A run passes the *conventional score* if its output is valid and every estimate and interval bound lies within 0.25 reference SE, which is what an execution-and-answer benchmark checks. *Consistency* compares a decision with Eq. (1) applied to the agent's own numbers. Intervals are percentile bootstraps over the 72 pairs (4,000 resamples), because the pair is the independent unit. The confirmatory comparisons (C1–C5 below) and the Holm adjustment were fixed in an analysis plan before the main run; everything else is exploratory.

5 Results

Preregistered comparisons of macro recall: structured vs. carry-forward (C1), vs. no tools (C2), oracle vs. structured (C3), narrative (C4) and implicit (C5) vs. structured. Full results: Table A1.

With computation and an explicit standard, most agents revise selectively. In the structured condition, seven of nine models reach a macro recall of at least 0.95 (range 0.95–1.00),

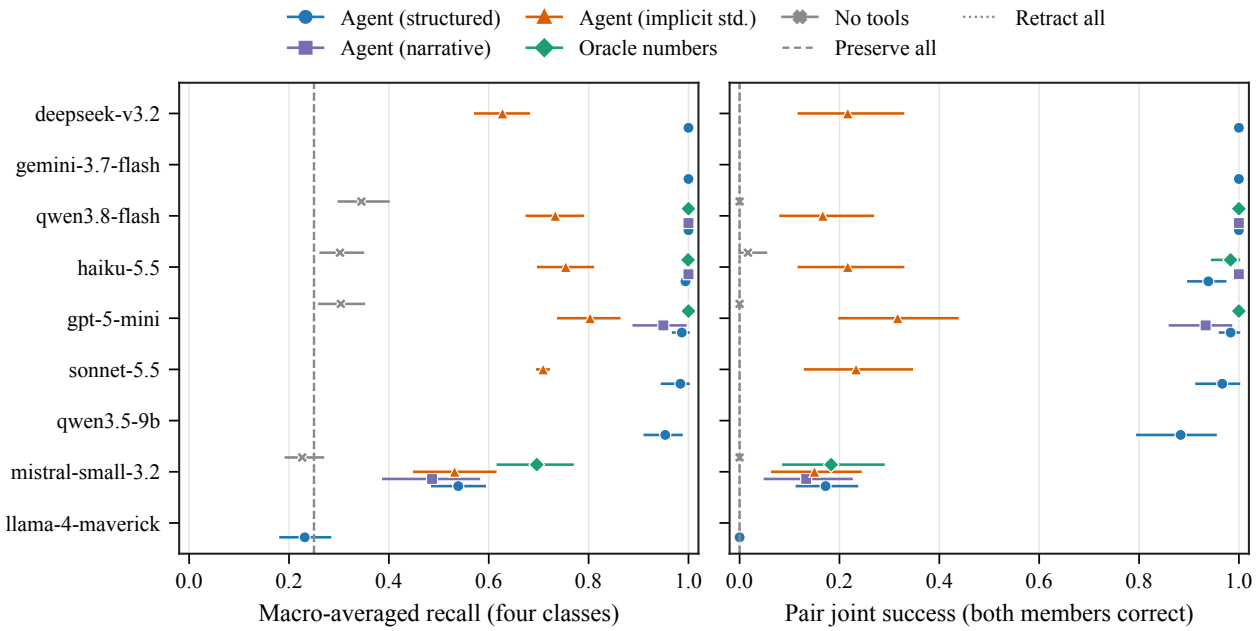


Figure 2 | Claim-revision performance by model and condition (points: estimates; bars: 95% pair-cluster bootstrap CIs). Left: macro recall over the four reference decisions. Right: share of matched A–E pairs with both members fully correct. Gray lines: deterministic baselines (preserve-all and retract-all both score 0.25 macro recall and 0 joint success).

while Mistral Small 3.2 reaches 0.54 and Llama 4 Maverick reaches 0.23 (Figure 2). The capable agents keep the claims that should survive, including in the robust-control family F, and change the claims that should change. Every model except Llama 4 Maverick beats carrying claims forward (C1; Holm $p = 0.013$, the smallest value attainable with 4,000 resamples and 27 tests; Llama: -0.02, $p = 1.000$). For these models, verified numbers add almost nothing: C3 is +0.01 [+0.00, +0.01] for Haiku 5.5 and +0.01 for GPT-5-mini. Removing the claim-to-estimand mapping costs little: C4 is +0.01 and -0.04. Across three repetitions, 97% (Haiku) and 99% (GPT-5-mini) of claim decisions were identical in every repetition.

Without computation, agents hedge rather than discriminate. The correction text does not reveal which pair member breaks a conclusion, and the no-tools condition confirms it. Macro recall falls to 0.23–0.34 (C2, structured minus no tools: +0.69 for Haiku, Holm $p = 0.013$), and joint pair success to 0.02. On breaking members, agents kept the focal claim 59% of the time and called it insufficient 28% of the time. On preserving members they revised it 44% of the time. They almost never chose qualify or retract (9% of claims).

Without the numeric standard, agents over-revise. Withholding Eq. (1), while still defining the categories in words, has the largest effect of any ablation (C5). The change in macro recall is -0.24 [-0.29, -0.19] for Haiku, -0.18 for GPT-5-mini, -0.27 for Qwen3.8 Flash, -0.27 for Sonnet and -0.37 for DeepSeek (all Holm $p = 0.013$). Mistral, already low, is unchanged (-0.01). The disagreements run in one direction (Figure A2 in Appendix A). Of 606 wrong decisions, 45% *qualify* a claim that still clears its threshold, and 21% retract or suspend a claim whose direction is established but whose magnitude fell below the claim. Only 3% defend a claim the standard would revise. Left to their own judgment, agents treat a smaller or less certain estimate as grounds for revision. They track how far the numbers moved rather than whether the evidence still meets the claim. The structured ceiling therefore reflects *applying* a given standard more than a calibrated default notion of support.

Weaker agents fail at different steps. Mistral Small 3.2 reaches 0.54 macro recall, and 0.70 even with verified numbers (C3 = +0.16). When its own numbers were correct, 27% of its decisions still contradicted the standard applied to them, and it revised 31% of the robust-control claims. Llama 4 Maverick often wrote code as text instead of calling the tool (valid-output rate 0.67).

Numerical and claim-level scores diverge where the claim step is skipped or done badly. The numbers-only pipeline passes the conventional score in 100% (Haiku) and 98% (GPT-5-mini) of scenarios. It also retains every unsupported claim and solves no matched pair (Figure A1, Appendix A). Among structured runs, 157 of the 1813 conventional passes still get a claim wrong, mostly through mapping errors in weaker models. Another 124 runs revise every claim correctly while failing the conventional score, usually because a number falls outside tolerance without crossing a decision boundary.

6 Failure Analysis

Table E1 (Appendix E) assigns each wrong decision to one cause: no output, no estimate, wrong numbers, or a decision contradicting the agent's own numbers. Three representative cases illustrate different failures. *Silent analysis on the wrong data* (GPT-5-mini, D, breaking member). The agent ran the script through Python's `exec`, which falls back to its default `data/original.csv`. It then reported the original estimates as corrected, applied the rule correctly to them, and preserved every claim. The claim error is purely an execution error, which only a numerical check exposes. *Over-revision against its own rule* (Claude Haiku 5.5, C, preserving member). The agent wrote "by the stated standard this is preserve (L>0 and t>=m)" (estimate 6.07, threshold 4), yet submitted qualify because "the original magnitude of 10.02 was materially overstated." This pattern accounts for 13 of Haiku's 15 structured-condition errors and dominates the implicit-standard condition. *Decisions without analysis* (Llama 4 Maverick). The agent narrated plans and wrote code as text instead of calling the tool. Of 144 runs, 47 never produced a valid submission and 67 submitted decisions without any successful code execution.

7 Limitations

Realism. The studies are short synthetic analyses with one correction and three claims each. Real errata involve several errors, ambiguous scope and interdependent claims, so ceiling performance here does not transfer to them. **The standard defines the labels.** Labels are correct relative to Eq. (1): Wald intervals, one threshold per claim, and no multiplicity adjustment. The implicit-standard condition therefore measures agreement with our standard, not absolute correctness. **Matching** in A–D holds the matched quantity equal by construction; other nuisance quantities, such as the change in the focal SE, are not matched. In E, the two members are different studies. **Coverage and power.** We test one prompt version and one temperature, and not every model was run in every condition. Per-family results rest on 12 pairs and are exploratory, and family E has only one breaking label. **Baselines.** Numbers-only carries claims forward by construction, and no-tools is uninformative by design; both measure default behavior, not achievable ceilings.

8 Conclusion

On 72 validated matched pairs, capable agents with code execution and an explicit evidential standard revise claims selectively and almost perfectly. Without execution they cannot see which corrections matter, without the numeric standard they over-revise supported claims, and conventional scoring fully credits a pipeline that never revises a claim. Evaluations of scientific agents should score the claims an agent asserts, not only the numbers it computes, and should state the standard those claims are held to.

References

Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. 1985. On the logic of theory change: Partial meet contraction and revision functions. *Journal of Symbolic Logic*, 50(2):510–530. https://doi.org/10.2307/2274239.

A. Colin Cameron and Douglas L. Miller. 2015. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372. https://doi.org/10.3368/jhr.50.2.317.

Jun Shern Chan, Neil Chowdhury, Oliver Jaffe, James Aung, Dane Sherburn, Evan Mays, et al. MLE-bench: Evaluating machine learning agents on machine learning engineering. In *International Conference on Learning Representations*, 2025.

Ziru Chen, Shijie Chen, Yuting Ning, Qianheng Zhang, Boshi Wang, Botao Yu, et al. ScienceAgentBench: Toward rigorous assessment of language agents for data-driven scientific discovery. In *International Conference on Learning Representations*, 2025.

Roi Cohen, Eden Biran, Ori Yoran, Amir Globerson, and Mor Geva. 2024. Evaluating the ripple effects of knowledge editing in language models. *Transactions of the Association for Computational Linguistics*, 12:283–298. https://doi.org/10.1162/tacl_a_00644.

Ferric C. Fang, R. Grant Steen, and Arturo Casadevall. 2012. Misconduct accounts for the majority of retracted scientific publications. *Proceedings of the National Academy of Sciences*, 109(42): 17028–17033. https://doi.org/10.1073/pnas.1212247109.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, et al. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11): 665–673. https://doi.org/10.1038/s42256-020-00257-z.

Andrew Gelman and Eric Loken. 2014. The statistical crisis in science. *American Scientist*, 102(6): 460–465.

Ken Gu, Ruoxi Shang, Ruien Jiang, Keying Kuang, Richard-John Lin, Donghe Lyu, et al. BLADE: Benchmarking language model agents for data-driven science. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 13936–13971, Miami, Florida, USA, 2024. Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-emnlp.815.

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. Annotation artifacts in natural language inference data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 107–112, New Orleans, Louisiana, 2018. Association for Computational Linguistics. https://doi.org/10.18653/v1/N18-2017.

Miguel A. Hernán and James M. Robins. *Causal Inference: What If*. Chapman & Hall/CRC, Boca Raton, 2020. https://miguelhernan.org/whatifbook.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, et al. Large language models cannot self-correct reasoning yet. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. https://openreview.net/forum?id=IkmD3fKBPQ.

Ryo Kamoi, Yusen Zhang, Nan Zhang, Jiawei Han, and Rui Zhang. 2024. When can LLMs actually correct their own mistakes? A critical survey of self-correction of LLMs. *Transactions of the Association for Computational Linguistics*, 12:1417–1440. https://doi.org/10.1162/tacl_a_00713.

Sayash Kapoor, Benedikt Stroebl, Zachary S. Siegel, Nitya Nadgir, and Arvind Narayanan. 2024. AI agents that matter. *arXiv preprint arXiv:2407.01502*.

Daniël Lakens. 2017. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362. https://doi.org/10.1177/1948550617697177.

Roderick Little and Donald Rubin. *Statistical Analysis with Missing Data*. Wiley Series in Probability and Statistics. Wiley, third edition, 2019. ISBN 9780470526798. https://doi.org/10.1002/9781119482260.

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, et al.

Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36, pages 46534–46594, 2023. https://doi.org/10.52202/075280-2019.

Bodhisattwa Prasad Majumder, Harshit Surana, Dhruv Agarwal, Bhavana Dalvi Mishra, Abhijeetsingh Meena, Aryan Prakhar, et al. DiscoveryBench: Towards data-driven discovery with large language models. In *International Conference on Learning Representations*, 2025. https://arxiv.org/abs/2407.01725.

Samuel Messick. 1995. Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist*, 50(9): 741–749. https://doi.org/10.1037/0003-066X.50.9.741.

Linlu Qiu, Fei Sha, Kelsey Allen, Yoon Kim, Tal Linzen, and Sjoerd van Steenkiste. 2026. Bayesian teaching enables probabilistic reasoning in large language models. *Nature Communications*, 17: 1238. https://doi.org/10.1038/s41467-025-67998-6.

Deborah Raji, Emily Denton, Emily M. Bender, Alex Hanna, and Amandalynne Paullada. AI and the everything in the whole wide world benchmark. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021. https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/084b6fbb10729ed4da8c3d3f5a3ae7c9-Abstract-round2.html.

Anka Reuel, Amelia Hardy, Chandler Smith, Max Lamparth, Malcolm Hardy, and Mykel J. Kochenderfer. BetterBench: Assessing AI benchmarks, uncovering issues, and establishing best practices. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 37, pages 21763–21813, 2024. https://doi.org/10.52202/079017-0685.

Zachary S. Siegel, Sayash Kapoor, Nitya Nadgir, Benedikt Stroebl, and Arvind Narayanan. 2024. CORE-Bench: Fostering the credibility of published research through a computational reproducibility agent benchmark. *arXiv preprint arXiv:2409.11363*.

Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. 2011. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366. https://doi.org/10.1177/0956797611417632.

Giulio Starace, Oliver Jaffe, Dane Sherburn, James Aung, Jun Shern Chan, Leon Maksin, et al. PaperBench: Evaluating AI's ability to replicate AI research. In *International Conference on Machine Learning*, 2025.

Ronald L. Wasserstein and Nicole A. Lazar. 2016. The ASA statement on p-values: Context, process, and purpose. *The American Statistician*, 70(2):129–133. https://doi.org/10.1080/00031305.2016.1154108.

Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, et al. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8541–8565, Miami, Florida, USA, 2024. Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-main.486.

Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, et al. 2025. Establishing best practices for building rigorous agentic benchmarks. *arXiv preprint arXiv:2507.02825*.

A Full Results, Coverage and Cost

Model	Condition	Macro recall [95% CI]	Acc.	Unsup. ret. ↓	Unnec. rev. ↓	Pair joint	Conv. pass	Valid
DeepSeek V3.2	Agent, structured	1.00 [1.00, 1.00]	1.00	0.00	0.00	1.00	0.99	1.00
	Agent, implicit standard	0.63 [0.57, 0.68]	0.76	0.00	0.17	0.22	0.99	1.00
Gemini 3.7 Flash	Agent, structured	1.00 [1.00, 1.00]	1.00	0.00	0.00	1.00	1.00	1.00
GPT-5-mini	Agent, structured	0.99 [0.97, 1.00]	1.00	0.02	0.00	0.98	0.97	1.00
	Agent, narrative	0.95 [0.89, 0.99]	0.99	0.06	0.00	0.93	0.91	1.00
	Agent, implicit standard	0.80 [0.74, 0.86]	0.84	0.02	0.14	0.32	0.98	1.00
	Oracle numbers	1.00 [1.00, 1.00]	1.00	0.00	0.00	1.00	0.00	1.00
	No tools	0.30 [0.26, 0.35]	0.57	0.51	0.33	0.00	0.03	1.00
	Numbers only (carry-forward)	0.25 [0.25, 0.25]	0.78	1.00	0.00	0.00	0.98	1.00
Claude Haiku 5.5	Agent, structured	0.99 [0.99, 1.00]	0.99	0.00	0.01	0.94	1.00	1.00
	Agent, narrative	1.00 [1.00, 1.00]	1.00	0.00	0.00	1.00	1.00	1.00
	Agent, implicit standard	0.75 [0.70, 0.81]	0.78	0.00	0.20	0.22	0.98	1.00
	Oracle numbers	1.00 [1.00, 1.00]	1.00	0.00	0.00	0.98	0.00	1.00
	No tools	0.30 [0.26, 0.35]	0.68	0.76	0.17	0.02	0.10	1.00
	Numbers only (carry-forward)	0.25 [0.25, 0.25]	0.78	1.00	0.00	0.00	1.00	1.00
Llama 4 Maverick	Agent, structured	0.23 [0.18, 0.28]	0.53	0.45	0.03	0.00	0.15	0.67
Mistral Small 3.2	Agent, structured	0.54 [0.49, 0.59]	0.67	0.20	0.27	0.17	0.55	0.99
	Agent, narrative	0.49 [0.39, 0.58]	0.67	0.33	0.24	0.13	0.37	0.99
	Agent, implicit standard	0.53 [0.45, 0.61]	0.64	0.16	0.30	0.15	0.55	1.00
	Oracle numbers	0.70 [0.62, 0.77]	0.64	0.01	0.38	0.18	0.00	1.00
	No tools	0.23 [0.19, 0.27]	0.58	0.72	0.24	0.00	0.02	0.98
	Numbers only (carry-forward)	0.24 [0.22, 0.25]	0.73	0.98	0.00	0.00	0.49	0.95
Qwen3.5-9B	Agent, structured	0.95 [0.91, 0.99]	0.95	0.00	0.00	0.88	0.92	0.95
Qwen3.8 Flash	Agent, structured	1.00 [1.00, 1.00]	1.00	0.00	0.00	1.00	1.00	1.00
	Agent, narrative	1.00 [1.00, 1.00]	1.00	0.00	0.00	1.00	0.99	1.00
	Agent, implicit standard	0.73 [0.68, 0.79]	0.78	0.00	0.19	0.17	1.00	1.00
	Oracle numbers	1.00 [1.00, 1.00]	1.00	0.00	0.00	1.00	0.00	1.00
	No tools	0.34 [0.30, 0.40]	0.73	0.72	0.12	0.00	0.06	1.00
	Numbers only (carry-forward)	0.25 [0.25, 0.25]	0.78	1.00	0.00	0.00	1.00	1.00
Claude Sonnet 5.5	Agent, structured	0.98 [0.95, 1.00]	0.99	0.00	0.00	0.97	0.99	0.99
	Agent, implicit standard	0.71 [0.70, 0.72]	0.79	0.00	0.16	0.23	1.00	1.00
Baseline	Preserve all	0.25 [0.25, 0.25]	0.78	1.00	0.00	0.00	–	1.00
Baseline	Retract all	0.25 [0.25, 0.25]	0.06	0.00	1.00	0.00	–	1.00

Table A1 | Main results, end to end (failed runs count as incorrect). Macro recall with 95% pair-cluster bootstrap CI over 72 pairs. Unsup. ret.: share of claims needing revision that were preserved. Unnec. rev.: share of claims that should be preserved that were changed. Pair joint: share of A–E pairs with both members fully correct. Conv. pass: valid output with all estimates within 0.25 SE. Models with three repetitions of the structured condition pool them.

B Validation Details

For every scenario, `claimrev.benchmark.validate` checks: (1) regeneration from (family, index, attempt) reproduces the original and corrected data byte for byte; (2) the agent-facing `original_analysis.py` run on `original.csv` matches both the stored reference and `original_results.json` to 10^{-9}; (3) the hidden corrected-analysis script run on `corrected.csv` matches the corrected reference; (4) every estimate and SE matches an independent implementation (`statsmodels` OLS with HC1 or cluster covariance; `CompareMeans` with unequal variances; `scipy.stats.sem`); (5) the corrected data differ from the original only in the logged records and field, with the stated kind of change (label flip, exact offset, or missing to observed); B adds exactly one column, and E leaves the data unchanged; (6) labels equal Eq. (1), all original labels are preserve, and every label is robust to ± 0.15 SE; (7) the README states every claim, threshold and original estimate, and the correction counts match the log; (8) agent-facing files contain no role or decision words outside the standard itself, no corrected interval bounds, and no hidden files. Per pair, the validator also checks that there are two members with the expected conditions and that the matching diagnostics fall within the predeclared tolerances. All 144 scenarios and 72 pairs pass. A test also corrupts a scenario (label, data and README) and confirms that the validator fails it.

Model	Condition	Runs	Valid	Tool calls	Tokens	Seconds	Cost ($)
DeepSeek V3.2	Agent, structured	144	1.00	10.8	68.0k	314	1.64
DeepSeek V3.2	Agent, implicit standard	144	1.00	11.9	81.3k	221	1.94
Gemini 3.7 Flash	Agent, structured	144	1.00	5.6	25.4k	30	4.38
Qwen3.8 Flash	Agent, structured	144	1.00	6.5	44.5k	244	0.65
Qwen3.8 Flash	Agent, narrative	144	1.00	6.6	41.2k	87	0.62
Qwen3.8 Flash	Agent, implicit standard	144	1.00	7.3	63.9k	150	1.01
Qwen3.8 Flash	Oracle numbers	144	1.00	1.0	3.2k	25	0.13
Qwen3.8 Flash	No tools	144	1.00	1.2	21.5k	146	0.87
Qwen3.8 Flash	Numbers only (carry-forward)	144	1.00	6.0	31.1k	57	0.41
Claude Haiku 5.5	Agent, structured	432	1.00	3.2	14.2k	35	0.95
Claude Haiku 5.5	Agent, narrative	144	1.00	3.9	17.1k	9	0.35
Claude Haiku 5.5	Agent, implicit standard	144	1.00	2.8	13.0k	11	0.32
Claude Haiku 5.5	Oracle numbers	144	1.00	1.0	3.3k	5	0.11
Claude Haiku 5.5	No tools	144	1.00	1.0	5.6k	12	0.24
Claude Haiku 5.5	Numbers only (carry-forward)	144	1.00	3.5	14.7k	7	0.28
GPT-5-mini	Agent, structured	432	1.00	3.4	9.4k	21	1.86
GPT-5-mini	Agent, narrative	144	1.00	3.6	9.4k	20	0.65
GPT-5-mini	Agent, implicit standard	144	1.00	3.4	9.2k	20	0.67
GPT-5-mini	Oracle numbers	144	1.00	1.0	2.1k	10	0.30
GPT-5-mini	No tools	144	1.00	1.0	3.1k	17	0.47
GPT-5-mini	Numbers only (carry-forward)	144	1.00	3.5	8.5k	13	0.48
Claude Sonnet 5.5	Agent, structured	144	0.99	3.6	16.3k	10	6.22
Claude Sonnet 5.5	Agent, implicit standard	144	1.00	3.7	16.5k	12	6.53
Qwen3.5-9B	Agent, structured	144	0.95	6.7	45.9k	231	0.63
Mistral Small 3.2	Agent, structured	432	0.99	4.6	20.9k	91	0.96
Mistral Small 3.2	Agent, narrative	144	0.99	4.7	17.9k	67	0.27
Mistral Small 3.2	Agent, implicit standard	144	1.00	4.7	20.8k	78	0.32
Mistral Small 3.2	Oracle numbers	144	1.00	1.0	3.0k	15	0.06
Mistral Small 3.2	No tools	144	0.98	1.0	11.1k	40	0.17
Mistral Small 3.2	Numbers only (carry-forward)	144	0.95	4.1	15.5k	76	0.25
Llama 4 Maverick	Agent, structured	144	0.67	2.0	129.6k	481	2.00

Table A2 | Run coverage and resource use (main tag; means per run, cost totals as reported by OpenRouter). Scored main runs: 5,328, $35.72. Recorded spend including the pilot (288 runs, $3.37, excluded from all analyses) and 220 duplicate runs from concurrent schedulers ($2.73, kept but not scored) was $41.83; OpenRouter billed slightly more because runs interrupted while being rescheduled left no record.

C Data-Generating Processes

All generators are deterministic in (family, pair index, attempt) through a SHA-256-seeded `numpy` generator. Exact parameters are stored with each scenario in `hidden/reference.json`.

A (subgroup reversal). $N \in [3600, 4600]$ records. About 30% belong to the minority population, and labels are positive with probability 0.4. For each record, the joint correctness of the new and baseline classifiers is drawn from a four-category distribution. In the majority population, only the new model is correct with probability 0.105–0.12 and only the baseline with 0.03–0.04; in the minority population the corresponding ranges are 0.135–0.15 and 0.03–0.04. A correction re-adjudicates k labels among records where only the new model was correct, which lowers every paired accuracy difference containing the record by $2/n$. The breaking member concentrates all k in the minority population; the preserving member spreads them proportionally across populations. Every claim uses $m = 0.04$.

B (confounding). A latent pre-exposure covariate z drives exposure, $\Pr(x = 1) = \sigma(-0.2 + 1.2z)$, and the outcome $y = 50 + \beta x + \gamma z + 0.15\,\text{age} + \varepsilon$. Here γ is set so that the original adjusted estimate lies near 4, and $\beta \in \{0, 1.1, 0.9\}$ with (n, σ) chosen for the retract, qualify and insufficient targets respectively. The breaking member reveals z. The preserving member reveals a prognostic covariate built from the original residuals and orthogonalized against the original regressors, scaled to give the same increase in R^2. Thresholds: $m = 2$ (unadjusted and adjusted effects) and 0.05 per year (age).

C (measurement). A 1:1 randomized trial with four devices. Device X serves 30% of participants, 90% of them treated; device Y serves an equal number, balanced across arms. In the breaking world readings from X carry an offset b, chosen so that the expected original

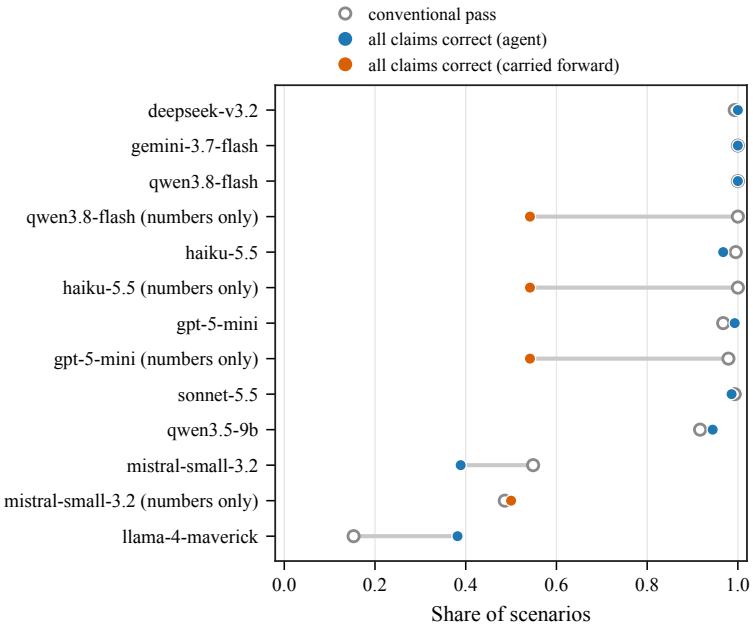


Figure A1 | Conventional pass rate (open) versus the share of scenarios with every claim correct (filled), structured condition and numbers-only pipelines. Numbers-only runs get full numerical credit but fail every scenario in which a claim should change (54% of scenarios are all-preserve).

effect falls in $[5.5, 7]$. The breaking correction removes b from X and the preserving correction removes the same b from Y. Thresholds: overall mean $m = 4$, effects $m = 3$.

D (missingness). A 1:1 trial with 35% of outcomes missing. Recovered outcomes are drawn from $\mathcal{N}(-6, 6^2)$ (non-responders). The breaking member assigns all k recovered values to treated participants and the preserving member splits the same values across arms in proportion to the missing counts. Thresholds: overall mean $m = 3$, effects $m = 2.5$.

E (inference). 24–36 clusters of about 38 participants, with treatment assigned by cluster. The breaking study has a cluster random intercept (SD 3, residual SD 7). The preserving study has no cluster effect and residual SD $\sqrt{58}$. After generation, the arm coefficient of the breaking study is shifted to 3.4 HC1 standard errors, and the preserving study is shifted so that all three original point estimates equal the breaking study's. The arm threshold is $m = 0.6\times$ the original estimate; the age and sex thresholds are 0.08 and 0.6.

F (controls). The A, C and D mechanisms, always spread proportionally or applied to the balanced device, at magnitude k (or b) and $3k$ (or $3b$). The focal claim must be preserve under both.

D Prompts and Output Schema

The system prompt for the structured agent condition (prompt version p1) reads:

> You are a scientific analysis agent working on a study that has received a correction. The study description, the original claims, the correction, and the evidential standard are in README.md (shown below); the data and the original analysis script are in your working directory. Your job: 1. Carry out the corrected analysis described in README.md. 2. For every original claim, decide whether it should be preserved, qualified, retracted, or marked insufficient_evidence, by applying the evidential standard to the corrected results. Give the revised wording of each claim. 3. Cite, for each decision, the run_id of the tool run whose output supports it. 4. Call submit with the corrected estimates for every estimand and a decision for every claim. [Tool and budget description.]

The `submit` schema requires `estimates` (estimand id, estimate, CI bounds) and `claims` (claim id, decision drawn from the four labels, revised claim, evidence reference, reasoning summary), with an optional free-text `uncertainty`. The prompts for the other conditions, the

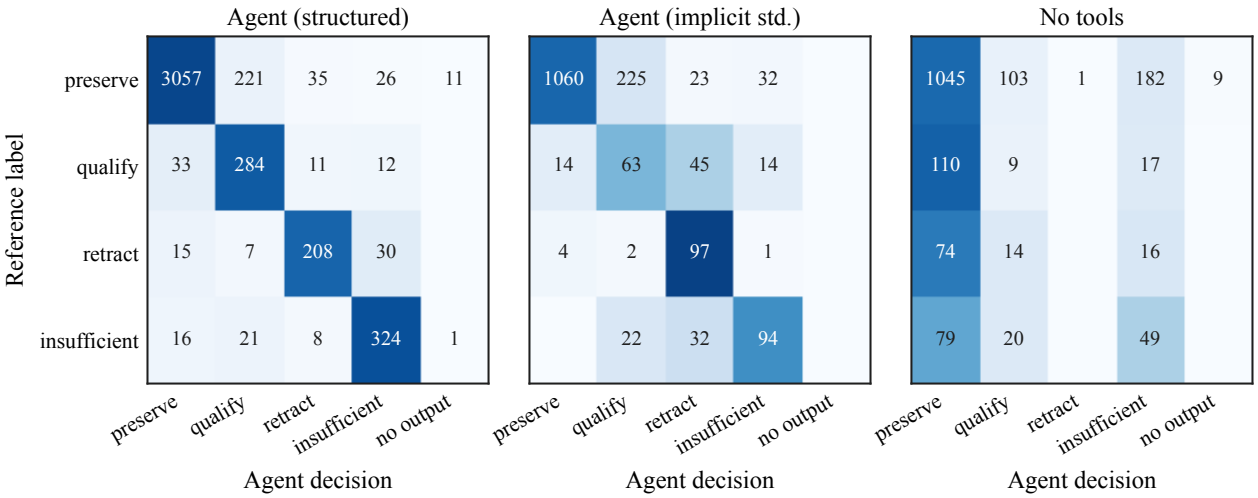


Figure A2 | Reference label (rows) versus decision (columns), pooled over the four models run in every condition and all repetitions. Shading: row share; numbers: claim counts.

README templates and the narrative and implicit variants are in `claimrev/agents/prompts.py` and `claimrev/benchmark/build.py`.

E Error Decomposition

Every incorrect claim decision in an agent condition falls into exactly one category. *No output*: the run produced no valid submission. *No estimate*: the claim was decided without a reported number for its estimand. *Wrong numbers*: the agent's own estimate or interval was more than 0.25 reference SE off, and the decision follows the standard applied to those wrong numbers. *Rule not followed*: the decision contradicts the standard applied to the agent's own numbers. In the implicit-standard condition, "rule not followed" measures disagreement with a standard the agent was not shown.

Model	Condition	Claims	Wrong	No output	No estimate	Wrong numbers	Rule not followed
DeepSeek V3.2	Agent, structured	432	0	0	0	0	0
DeepSeek V3.2	Agent, implicit standard	432	103	0	0	1	102
Gemini 3.7 Flash	Agent, structured	432	0	0	0	0	0
Qwen3.8 Flash	Agent, structured	432	0	0	0	0	0
Qwen3.8 Flash	Oracle numbers	432	0	0	0	0	0
Qwen3.8 Flash	No tools	432	118	0	14	97	7
Qwen3.8 Flash	Agent, implicit standard	432	95	0	0	0	95
Claude Haiku 5.5	Agent, structured	1296	15	0	0	0	15
Claude Haiku 5.5	Oracle numbers	432	1	0	1	0	0
Claude Haiku 5.5	No tools	432	140	0	2	83	55
Claude Haiku 5.5	Agent, implicit standard	432	95	0	0	0	95
GPT-5-mini	Agent, structured	1296	5	0	0	5	0
GPT-5-mini	Oracle numbers	432	0	0	0	0	0
GPT-5-mini	No tools	432	186	0	95	75	16
GPT-5-mini	Agent, implicit standard	432	70	0	0	2	68
Claude Sonnet 5.5	Agent, structured	432	4	3	0	0	1
Claude Sonnet 5.5	Agent, implicit standard	432	89	0	0	0	89
Qwen3.5-9B	Agent, structured	432	22	21	0	1	0
Mistral Small 3.2	Agent, structured	1296	427	12	0	93	322
Mistral Small 3.2	Oracle numbers	432	156	0	156	0	0
Mistral Small 3.2	No tools	432	181	9	71	50	51
Mistral Small 3.2	Agent, implicit standard	432	154	0	0	25	129
Llama 4 Maverick	Agent, structured	432	203	141	0	50	12

Table E1 | Decomposition of incorrect claim decisions (all repetitions). Wrong numbers and rule not followed apply only to runs that reported an estimate.